Day 17 Problem Set

Geometry Summer Credit | Angle and Arc Relationships

Name:	Date:	Period:

Directions

For each problem, identify the vertex location and intercepted arc before writing an equation.

Part A: Central Angles and Arc Measures

1.	Central angle AOB measures 64 degrees. Find minor arc AB.

2.	Minor arc CD measures 132 degrees. Find central angle COD.

3.	Minor arc EF measures 98 degrees. Find major arc EF.

4.	A diameter divides a circle into two arcs. Find each arc measure.

5.	A circle has consecutive arcs of 75 degrees, 110 degrees, 95 degrees, and x degrees. Find x.

6.	Three central angles are x, x+30, and 2x+10. Find x and all three arc measures.

7.	Central angle AOB is 5x+10 degrees and intercepted arc AB is 8x-44 degrees. Find x and the common measure.

Part B: Inscribed Angles

8.	An inscribed angle intercepts a 100-degree arc. Find the angle.

9.	An inscribed angle measures 42 degrees. Find its intercepted arc.

10.	An inscribed angle intercepts a 154-degree arc. Find the angle.

11.	An inscribed angle measures 71 degrees. Find its intercepted arc.

12.	An inscribed angle intercepts a semicircle. Find the angle.

Day 17 Problem Set

Continued | Mixed Relationships and Algebra

13.	Inscribed angles ACB and ADB intercept the same arc AB. If angle ACB=47 degrees, find angle ADB.

14.	A central angle and an inscribed angle intercept the same 136-degree arc. Find both angle measures.

15.	An inscribed angle is 3x+4 degrees and its intercepted arc is 8x-16 degrees. Find x and the angle measure.

16.	A central angle is 7x-5 degrees and an inscribed angle intercepting the same arc is 3x+5 degrees. Find x and both angles.

17.	Two inscribed angles intercept the same arc. Their measures are 5x-8 and 3x+18. Find x and both angles.

18.	Minor arc AB is 4x+20 degrees and major arc AB is 8x+40 degrees. Find x and both arc measures.

19.	Error analysis: A student says an inscribed angle intercepting a 120-degree arc measures 120 degrees. Explain and correct the error.

20.	Error analysis: A student doubles a 70-degree central angle to find its intercepted arc. Explain and correct the error.

21.	Spiral review: State the center and radius of (x-2)^2+(y+4)^2=36.

22.	Challenge: Points A and B are endpoints of a diameter. Point C is on the circle. Explain why angle ACB must be a right angle.